

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1892

Loudon 345 kV
September 20, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1892. The Interconnection Customer proposes a 264.0 MW solar photovoltaic facility interconnecting at the Loudon 345 kV substation in the Virginia area. The request was received on September 29, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 464 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,796,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

CLAYTONIA - SCURRY 138 KV: 121.7% loading (198 MVA) following the outage of the LOUDON - THOMASTON 345 KV; DFAX = 13.1%.

The WHEELER - MAMMOTH PACIFIC 2 138 KV line is overloaded to 111.0% of its 407 MVA emergency rating for the loss of the LOUDON - THOMASTON 345 KV. The project contributes 29.6% DFAX (78.2 MW) to this overload.

SCURRY - LEESBURG 138 KV: 110.9% loading (180 MVA) following the outage of the LOUDON - THOMASTON 345 KV; DFAX = 13.1%.

The MAMMOTH PACIFIC 2 - T.S. POWER 138 KV line is overloaded to 109.7% of its 442 MVA emergency rating for the loss of the LOUDON - THOMASTON 345 KV line. The project contributes 29.6% DFAX (78.2 MW) to this overload.

The WHEELER - PASCO CNTY SOLID WASTE RESOURCE RECOVERY 138 KV is overloaded to 109.2% of its 649 MVA emergency rating for the loss of the LOUDON - THOMASTON 345 KV line. The project contributes 18.3% DFAX (48.4 MW) to this overload.

TWIN LAKES ALTE - T.S. POWER 138 KV: 108.8% loading (468 MVA) following the outage of the LOUDON - THOMASTON SUBSTATION 345 KV Line; DFAX = 29.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CLAYTONIA - SCURRY 138 KV	Rebuild line with higher-capacity conductor	\$8,506,000	39% (\$3,283,000)
WHEELER - MAMMOTH PACIFIC 2 138 KV line	Rebuild line with higher-capacity conductor	\$4,710,000	18% (\$860,000)
LESBURG - SCURRY 138 KV	Rebuild line with higher-capacity conductor	\$4,392,000	39% (\$1,695,000)
MAMMOTH PACIFIC 2 - T.S. POWER 138 KV	Reconductor line	\$1,427,000	18% (\$261,000)
WHEELER - PASCO CNTY SOLID WASTE RESOURCE RECOVERY 138 KV line	Reconductor line	\$1,561,000	20% (\$315,000)
TWIN LAKES ALTE - T.S. POWER 138 KV	Reconductor line	\$2,091,000	18% (\$382,000)
	Total allocated to project		\$6,796,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.